CLASS 10 · ICSE · SAMPLE PAPER

MathSetu Sample Paper 1: Mathematics

Time allowed: 2 hours 30 minutes · Maximum marks: 80

GENERAL INSTRUCTIONS

1. The blue number beside each heading is the question number. Questions 1 to 3 make up Section A (40 marks) and Questions 4 to 10 make up Section B (40 marks).
2. Attempt all questions from Section A and any four questions from Section B.
3. The parts of the questions are numbered 1 to 42 straight through, and the marks for each part are shown beside it.
4. All working, including rough work, must be shown clearly on the same sheet as the rest of the answer. Marks are lost if essential working is left out.
5. Mathematical tables may be used. Take $\pi = \frac{22}{7}$ unless another value is given.

1 Section A, Question 1: choose the correct answer (15 marks)

1. An article is marked at ₹800 and sold within the same state. If the rate of GST is 18%, the CGST charged is: (a) ₹36 (b) ₹72 (c) ₹80 (d) ₹144 1 mark

2. ₹500 is deposited every month for 12 months in a recurring deposit account. The equivalent principal for one month, used to calculate the interest, is: (a) ₹6000 (b) ₹33,000 (c) ₹39,000 (d) ₹78,000 1 mark

3. A ₹100 share is bought at ₹125 and pays a dividend of 10%. The rate of return on the investment is: (a) 8% (b) 10% (c) 12.5% (d) 25% 1 mark

4. The solution set of $-3 < 2x - 1 \leq 5$, where x is a whole number, is: (a) $\{1, 2, 3\}$ (b) $\{0, 1, 2\}$ (c) $\{0, 1, 2, 3\}$ (d) $\{-1, 0, 1, 2, 3\}$ 1 mark

5. The roots of $x^2 - 6x + 9 = 0$ are: (a) real and unequal (b) real and equal (c) not real (d) equal in size and opposite in sign 1 mark

6. The mean proportional between 4 and 25 is: (a) 10 (b) 14.5 (c) 21 (d) 100 1 mark

7. The remainder when $x^3 - 2x^2 + 3x - 4$ is divided by $x - 2$ is: (a) -4 (b) 0 (c) 2 (d) 6 1 mark

8. A is a matrix of order 2×3 and B is a matrix of order 3×1. The order of AB is: (a) 3×3 (b) 2×1 (c) 1×2 (d) AB is not possible 1 mark

9. The nth term of an A.P. is $5n - 2$. Its common difference is: (a) -2 (b) 2 (c) 3 (d) 5 1 mark

10. The 5th term of the G.P. $3, 6, 12, \ldots$ is: (a) 24 (b) 36 (c) 48 (d) 96 1 mark

11. The image of the point $(3, -4)$ after reflection in the y-axis is: (a) $(3, 4)$ (b) $(-3, 4)$ (c) $(-3, -4)$ (d) $(-4, 3)$ 1 mark

12. The slope of the line $3x - 4y + 7 = 0$ is: (a) $\frac{3}{4}$ (b) $-\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{7}{4}$ 1 mark

13. An arc of a circle subtends an angle of 110° at the centre. The angle it subtends at a point on the remaining part of the circle is: (a) 55° (b) 70° (c) 110° (d) 220° 1 mark

14. The volume of a cone of radius 3 cm and height 7 cm is: (a) 22 cm^3 (b) 66 cm^3 (c) 132 cm^3 (d) 198 cm^3 1 mark

15. A letter is chosen at random from the letters of the word TRIANGLE. The probability that it is a vowel is: (a) $\frac{1}{4}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{5}{8}$ 1 mark

2 Section A, Question 2 (12 marks)

16. Meera deposits ₹1200 every month for 2 years in a recurring deposit account that pays simple interest at 8% per annum. Find the interest she earns and the amount she gets on maturity. 4 marks

17. If $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$, find $A^2 - 2B$. 4 marks

18. Prove that $\frac{\cos A}{1 - \sin A} + \frac{\cos A}{1 + \sin A} = 2 \sec A$. 4 marks

3 Section A, Question 3 (13 marks)

19. A solid metal sphere of radius 6 cm is melted and recast into a solid cylinder of radius 4 cm. Find the height of the cylinder and its curved surface area, correct to one decimal place. 4 marks

20. $A(2, 3)$ and $B(6, -5)$ are two points. Find (i) the slope of AB (ii) the equation of the perpendicular bisector of AB (iii) the point where this bisector meets the x-axis. 4 marks

21. Use graph paper for this question, with 1 cm $=$ 1 unit on both axes. Plot the point $A(4, 3)$. Reflect A in the x-axis to get B, reflect A in the origin to get C, and reflect A in the y-axis to get D. (i) Write the coordinates of B, C and D. (ii) Name the figure $ABCD$. (iii) Find its perimeter and its area. 5 marks

4 Section B, Question 4 (10 marks)

22. A dealer in Delhi sells a washing machine, marked at ₹20,000, to a customer in Delhi at a discount of 10%. The rate of GST is 18%. Find the CGST, the SGST and the total amount the customer pays. 3 marks

23. Solve $x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. 3 marks

24. Find the mean of the following distribution by the step-deviation method. 4 marks

Class	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
Frequency	6	10	16	12	6

5 Section B, Question 5 (10 marks)

25. A man invests ₹9600 in ₹100 shares at a premium of 20%. The company declares a dividend of 15%. Find (i) the number of shares he buys (ii) his annual dividend (iii) the rate of return on his investment. 3 marks

26. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 4)^\circ$, $\angle B = (x + 40)^\circ$ and $\angle C = (3x + 16)^\circ$. Find x and all four angles. 3 marks

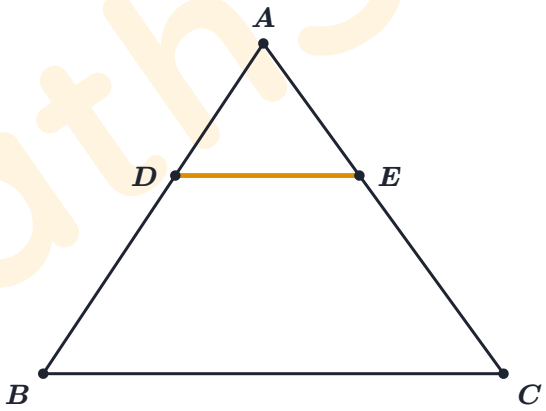
27. Use the Factor Theorem to show that $x - 2$ is a factor of $2x^3 + x^2 - 13x + 6$. Hence factorise the expression completely. 4 marks

6 Section B, Question 6 (10 marks)

28. Solve $3x - 7 \leq 7x + 5 < 19$, where x is an integer, and show the solution set on a number line. 3 marks

29. The 4th term of an A.P. is 22 and its 15th term is 66. Find the first term, the common difference and the sum of its first 8 terms. 3 marks

30. In $\triangle ABC$, D lies on AB and E lies on AC so that $DE \parallel BC$ and $AD : DB = 2 : 3$. (i) Prove that $\triangle ADE \sim \triangle ABC$. (ii) If $DE = 4$ cm, find BC. (iii) Find the ratio of the area of $\triangle ADE$ to the area of trapezium $DBCE$. 4 marks



7 Section B, Question 7 (10 marks)

31. What number must be added to each of 2, 6, 8 and 18 so that the four results are in proportion? 3 marks

32. The 3rd term of a G.P. is 12 and its 6th term is 96. Find the first term, the common ratio and the sum of its first 8 terms. 3 marks

33. From a point A on level ground, the angle of elevation of the top of a tower is 30°. On walking 40 m towards the tower to a point B, the angle of elevation becomes 60°. Find the height of the tower, correct to two decimal places, and the distance of B from the tower. (Take $\sqrt{3} = 1.732$.) 4 marks

8 Section B, Question 8 (10 marks)

34. Cards numbered 1 to 30 are put in a box and one card is drawn at random. Find the probability that the number on it is (i) a multiple of 4 (ii) a prime number (iii) a perfect square. 3 marks

35. In what ratio does the y-axis divide the line segment joining $A(-4, 1)$ and $B(6, 6)$? Also find the point of division. 3 marks

36. A conical tent is 12 m high and the radius of its base is 9 m. Find (i) the cost of the canvas for its curved surface at ₹70 per m^2 (ii) the volume of air inside it, correct to the nearest m^3. 4 marks

9 Section B, Question 9 (10 marks)

37. The length of a rectangular garden is 4 m more than its breadth, and its area is 96 m^2. Find its length, its breadth and its perimeter. 3 marks

38. (i) Chords AB and CD of a circle meet at a point P inside the circle. If $AP = 6$ cm, $PB = 4$ cm and $CP = 3$ cm, find CD. (ii) From a point T outside another circle, a tangent TQ and a secant TRS are drawn, with $TR = 4$ cm and $RS = 5$ cm. Find TQ. 3 marks

39. The marks of 11 students in a test are 12, 18, 25, 9, 30, 21, 15, 27, 33, 6 and 24. Find (i) the median (ii) the lower quartile (iii) the upper quartile (iv) the interquartile range. 4 marks

10 Section B, Question 10 (10 marks)

40. Find the equation of the line that passes through $(2, -1)$ and is parallel to the line $3x + 4y = 7$. Also write its y-intercept. 3 marks

41. Find x and y if $\begin{bmatrix} 2 & x \\ y & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$. 3 marks

42. Using a ruler and compasses only, construct $\triangle ABC$ with $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$. Construct the incircle of the triangle and measure its radius. 4 marks

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